Gowin FPGA Junction Temperature Manual

About This Guide

This manual describes Gowin FPGA products current management desgin and computing method for reference.

Before reading this manual, please refer to *SUG282, Gowin Power Analyzer User Guide*.

Table 1 shows the abbreviations and terminology used in this manual.

Table 1 Abbreviations and Terminology

Abbreviations and Terminology	Full Name
T_J	Junction Temperature of the Device
T_C	Case Temperature of the Device
T_A	Ambient Temperature
Θ_{CS}	Thermal resistance of case-to-heatsink
Θ_{SA}	Thermal resistance of heatsink-to-ambient
Θ_{JA}	Thermal resistance of junction-to-ambient
Θ_{JC}	Thermal resistance of junction-to-case

Introduction

Power Thermal Management is an important part of FPGA system design. In order to properly evaluate the thermal characteristics of the system design, system design or FPGA device selection is required based on the maximum allowable junction temperature specified in the FPGA device thermal characteristics data sheet.

The system designer should fully consider and analyze the current of the design module to ensure that the device and package do not exceed the junction temperature requirements.

There are many factors that affect the operating current of the FPGA.

For example: DIE size, heat sink size, airflow, power, PCB design, proximity to other devices, and user applications.

System designers should use power analysis tools to measure power consumption and analyze each switch characteristic. For further detailed information, please refer to *SUG282, Gowin Power Analyzer User Guide*. If the calculated T_{JMAX} exceeds the specified range (as shown in Table 2), please refer to the following to reduce the overall current and package temperature.

Power Analysis

Since the current is the most critical factor affecting the heat of the FPGA, excessive current can increase the heat of the FPGA and is likely to cause the partial load and the temperature of the power supply to be too high. Therefore, the key to solve the power consumption is to reduce the power consumption of the FPGA.

The total current of the entire FPGA design consists of three parts:

- Chip quiescent current
- Design quiescent current
- Design dynamic current

Chip Quiescent Current

When the FPGA is not configured after power-on, the chip quiescent current is mainly the power consumption of the leakage current of the transistor. It also includes the inrush current that is formed when the initial logic state of the transistor is indeterminate.

Design Quiescent Current

When the FPGA configuration is complete and when the design has not yet started, the design quiescent current refers to the quiescent current of the I/O that need to be maintained, clock management, and static power consumption of other parts of the circuit.

Design Dynamic Current

The design dynamic current refers to the power consumption of the design after the normal startup of the FPGA design;

This type of power consumption depends primarily on the level used by the chip and the occupancy of the FPGA's internal logic and routing resources.

The first part of the current depends on the FPGA chip hardware design.

The second and third part of the current is closely related to the user's design and FPGA settings. Here is the third part that needs to be optimized:

Designing dynamic currents account for about 90% of the total current, so reducing the design dynamic current is a key factor in reducing the overall system current.

1. The relationship between heat and power is as shown in the formula below:

   $T_J=T_A+ \text{Power} \times \Theta_{JA}$

   or:

   $T_J=T_A+\text{Power} \times (\Theta_{JC}+\Theta_{CS}+\Theta_{SA})$

2. Ways to reduce junction temperature:

   Increase airflow to reduce the case or ambient temperature.

3. The current can be reduced by:

   a) PCB scheme optimization
   b) Via holes added on PCB to improve air flow
   c) Separate the heat dissipation of Power ground, FPGA ground, other devices ground
   d) Disperse high temperature devices
   e) LDO consumes large energy and can be replaced by a two-stage DCDC scheme
   f) Optimize the clock design
   g) Reduce the logic flip rate when ensuring performance
   h) Prohibit unnecessary clock flipping
   i) Close unused logical regions
   j) Choose the appropriate heatsink and heatsink material: aluminum heatsink

Low Power System Design

FPGA dynamic power consumption is mainly reflected in the power consumption of memory, internal logic, clock, and I/O.

1. Software Algorithm Optimization
   In system design, clock flipping and user logic resource usage have a major impact on power consumption. It's suggested to reduce user logic resource usage, plan the design, and recude clock flipping.
2. Optimization of Resource Use Efficiency
   The power consumption of DSP, BRAM, IO, resources, etc. is relatively large. For detailed configuration, please refer to *SUG282, Gowin Power Analyzer User Guide*.
3. PCB Design
   When designing a PCB, the main idea is heat transfer and thermal isolation.
   - Select the BGA package as much as possible for chip selection
   - The EPAD is fully grounded to improve heat dissipation

- Make via holes for heat dissipation
- High-temperature devices such as power supplies are separated as much as possible
- Temperature sensitive devices should be isolated
- Design multiple layers
- The main path for signal transmission is as wide as possible

Heatsink Design

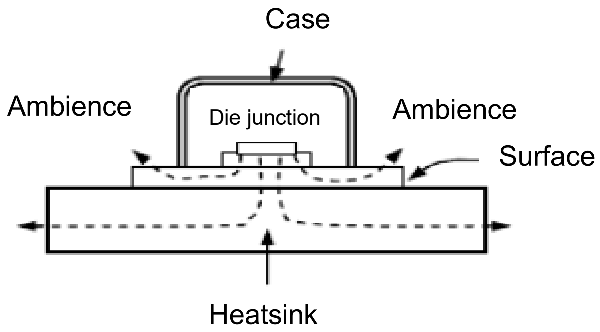
1. Parameter Definition

The definition of heatsink parameters is as shown in Table 2.

Table 2 Parameter Definition

Parameters	Description	Unit
Θ_{JA}	Thermal resistance of junction-to-ambient	℃/W
Θ_{JC}	Thermal resistance of junction-to-case	℃/W
Θ_{CS}	Thermal resistance of case-to-heatsink	℃/W
Θ_{SA}	Thermal resistance of heatsink-to-ambient	℃/W
T_J	Junction Temperature of semiconductor devices	℃
T_C	Case Temperature of Semiconductor Devices	℃
T_S	Heatsink temperature	℃
T_A	Ambient Temperature	℃
P_D	Service Power of Semiconductor Devices	W
ΔT_{SA}	Heatsink temperature rising	℃

2. Introduction of Thermal Contact Points between Heatsink and Chip



3. Calculating method of heatsink:

   Please refer to the formula:

   $\boldsymbol{\Theta_{SA}} = (\boldsymbol{T_J}\text{-}\ T_A)\ /\ P_D\ \text{-}\ \Theta_{JC} - \Theta_{CS}$

   Note!

   - Heatsink thermal resistance Θ_{SA} is the main basis for selecting a heatsink
   - T_J and Θ_{JC} are the parameters provided by the chip manual
   - P_D is a parameter required by the design
   - Θ_{CS} can be checked from the thermal design professional books

4. Detailed calculation process:

   a) Calculate total thermal resistance Θ_{JA}:

      $\Theta_{JA} = (T_{JMAX}\text{-}\ T_A)\ /\ P_D$

   b) Calculate the thermal resistance of heatsink Θ_{SA} or temperature rise ΔT_{SA}:

      - $\Theta_{SA} = (\Theta_{JA}\ \text{-}\ )\ /\ \Theta_{JC}\ \text{-}\ \Theta_{CS}$
      - $\Delta T_{SA} = \Theta_{SA} \times P_D$

   c) Determine the heatsink

      - Calculate the temperature rise of the heatsink ΔT_{SA} from the above formula, and then calculate the comprehensive heat transfer coefficient α of the heatsink
      - α = $7.2\Psi_1\Psi_2\Psi_3\{\sqrt{\sqrt{[(T_S - T_A)\ /\ 20]}}\ \}$
      - Ψ_1- Describe the effect of the heatsink l/b on α, (l is the length of the heatsink, b is the fin spacing)
      - Ψ_2 - Describe the effect of the heatsink h/b on α, (h is the height of the heatsink fin)
      - Ψ_3 - Describe the effect of the increase of width size W of the heatsink on α
      - $\sqrt{\sqrt{[(T_S - T_A)\ /\ 20]}}$ - Describe the effect of the maximum temperature of the surface of the heatsink on the ambient temperature rise to α

   d) Calculate the power dissipated between the two fins of the heatsink: Q_0

      - Q_0 =α × ΔT_{SA} ×（2h+b）×l

   e) Calculate the heat dissipation power $(P_D{}')$ according to the number of fins of the single-sided or double-sided heatsinks with ribs (n):
      Single-sided fin: $P_D{}'$=nQ_0

      - Single-sided fin: $P_D{}'$=nQ_0

- Double-sided fin: P_D'=2nQ_0
- If $P_D' > P_D$, it meets the requirements

Table 3 Thermal Resistance Information of Different Packages

Device	Package	Size	Pin Quantity	Θ_{JA}°C/W	Θ_{JC}°C/W	Note
EQ144	eLQFP	20x20	144	31.35	8.35	GW2A-18/GW2AR-18
EQ176	eLQFP	20x20	176	40.39	11.75	GW1N-9/GW2AR-18
FPG676	FCPBGA	27x27	676	12.222	0.067	GW5AT-138
LQ100	LQFP	14x14	100	49.9	9.88	GW1N-1/GW1N-2/ GW1N-1P5/GW1N-4/ GW1N-9
MG100P	MBGA	5x5	100	49.205	13.104	GW1NR-9
MG132P	MBGA	8x8	132	42.809	15.064	GW5AT-15
MG132X	MBGA	8x8	132	71.988	25.380	GW1N-2
MG160	MBGA	8x8	160	49.679	15.724	GW1N-4
MG196	MBGA	8x8	196	34.69	8.09	GW2A-18
MG196	MBGA	8x8	196	42.055	11.370	GW1N-9
PG256	PBGA	17x17	256	39.00	11.57	GW1N-9/GW1N-4
PG256S	PBGA	17x17	256	31.508	7.605	GW2A-18
PG484	PBGA	23x23	484	18.71	5.22	GW2A-55
PG484	PBGA	23x23	484	23.84	4.98	GW2A-18
PG484A	PBGA	23x23	484	17.153	3.334	GW5AT-138
QN32	QFN	5x5	32	45.61	14.5333	GW1N-2/GW1N-4
QN48	QFN	6x6	48	41.34	6.3463	GW1N-1/GW1N-2/GW1N-4/ GW1N-9
QN88	QFN	10x10	88	32.62	9.721	GW1N-2/GW1N-4/GW1N-9
QN88	QFN	10x10	88	29.84	7.25	GW2A-18/GW2AR-18
UG225	UBGA	13x13	225	18.85	5.46	GW5AT-60
UG256	UBGA	14x14	256	39.690	11.573	GW1N-9
UG256C	UBGA	14x14	256	26.88	10.49	GW5A-25
UG324	UBGA	15x15	324	24.408	5.279	GW2A-55
UG324	UBGA	15x15	324	32.37	9.48	GW2A-18
UG324A	UBGA	15x15	324	21.053	3.901	GW5AT-138
UG332	UBGA	17x17	332	37.990	11.393	GW1N-9
UG400	UBGA	17x17	400	31.571	9.504	GW2AN-9X/GW2AN-18X
UG484	UBGA	19x19	484	31.122	9.484	GW2A-18/GW2AN-9X/ GW2AN-18X
UG676	UBGA	21x21	676	18.927	5.687	GW2AN-55

Note!

- The above data is all set in the JEDEC standard environment. The Θ_{JA} and Θ_{JC} data of all packages were measured under the 2S2P test board, except for the

MG196 and PG484 packages of GW2A-18 devices whose Θ_{JC} data were measured under the 1S0P test board.

- The package thermal resistance of integrated circuits is related to chip size, frame structure, frame base island size, thermal conductivity of direct materials, design of PCB board, installation of external heatsinks, wind speed of product working environment and product working power. The above data is for reference only.

Customer Case

1. Background
   LED display is controlled via Ethernet. It adopts DC power supply 5V-to-1V, PCB size L=77MM, W=20MM, H=2MM, Ethernet working frequency 125MHz*8, sealed environment, and room temperature 70°C. FPGA adopts GW1N-QN88 device.
2. Case Description
   LED display is blurred with unstable Ethernet data.
3. Issue Description
   Through investigation, the problem is initially located at a high temperature of part of the power supply of the board caused by the high current of the FPGA at high temperature, and the high temperature of conduction causes the Ethernet chip not to work. The FPGA current consumption increases with the increase of ambient temperature and device case temperature. The ambient temperature is 70 °C and the surface temperature of the board reaches 110 °C. The FPGA chip is still working normally when the junction temperature exceeds 125 °C.
4. First Optimization
   Optimize the internal clock, and the current is significantly reduced after the function module is optimized, but the board temperature is still high, the FPGA consumes a large current, and the Ethernet chip is unstable.
5. Second Optimization
   Through the method mentioned in the above heatsink design section, the final option is to install an aluminum heatsink. The size is L=48MM, W=25MM, H=5MM, and the main FPGA chip and Ethernet chip are cooled. After the test, the current is further reduced after optimizing the software, and the temperature of the entire board is lowered, which has met the needs of customers.
6. Post Optimization Suggestion
   When designing PCB, it is suggested to separate the heat dissipation of FPGA, power supply ground, and Ethernet ground, so that the heat conduction of chip is small, the heat dissipation of circuit board is better, and the current of FPGA can be lower.

Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

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Revision History

Date	Version	Description
01/08/2019	1.0E	Initial version published.
09/01/2023	1.0.1E	The “Table 3 Thermal Resistance Information of Different Packages” in “Low Power System Design” updated.
02/28/2025	1.0.2E	The “Table 3 Thermal Resistance Information of Different Packages” in “Low Power System Design” updated.
04/11/2025	1.0.3E	The information of UG225 package in “Table 3 Thermal Resistance Information of Different Packages” in “Low Power System Design” updated.